

Continuous Distributions

classmate

Normal distⁿ, exponential distⁿ, Weibul distⁿ, gamma distⁿ

Normal Distribution

A continuous random variable x is said to have a normal distribution if its probability density function is given by

~~Probability density function~~

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} \exp \left\{ -\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2 \right\}$$

$$f(x) = \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2}, -\infty < x < \infty$$

$\sigma > 0$.

Here μ and σ are called parameters of normal distribution

μ = mean

σ = standard deviation.

Note: (i) Normal distribution is a limiting case of binomial distribution under the following conditions.

- (i) n , the no. of trials is indefinitely large i.e. $n \rightarrow \infty$, and
- (ii) neither p nor q is very small.

(ii) Normal distribution is a limiting case of poisson distribution, when its mean is large i.e. $\lambda \rightarrow \infty$

classmate
Date _____
Page _____

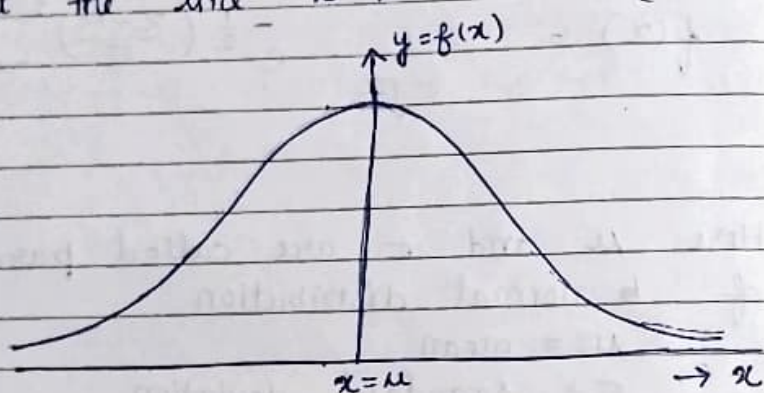
characteristics of the Normal distribution:-

The normal probability curve with mean μ and S.D σ is given by the equation

$$f(x) = \frac{1}{\sigma\sqrt{2\pi}} e^{-\frac{1}{2}\left(\frac{x-\mu}{\sigma}\right)^2}, -\infty < x < \infty.$$

has following probabilities:-

- (i) The curve is bell shaped and symmetrical about the line $x = \mu$ ($\mu = \text{mean}$)



Normal Probability curve

- (ii) Mean, median and mode of the distribution coincide. i.e. distribution is symmetrical.

- (iii) As x increases numerically, $f(x)$ decreases rapidly. The max. probability occurs at the point $x = \mu$ and given by

$$[f(x)]_{\max} = \frac{1}{\sigma\sqrt{2\pi}}$$

- (iv) Since $f(x)$ being the probability, can never be negative i.e. no portion of the curve lies below the x axis.

$$f''(x) = -\frac{f(x)}{\sigma^2} \left(1 - \frac{(x-\mu)^2}{\sigma^2} \right) = 0$$

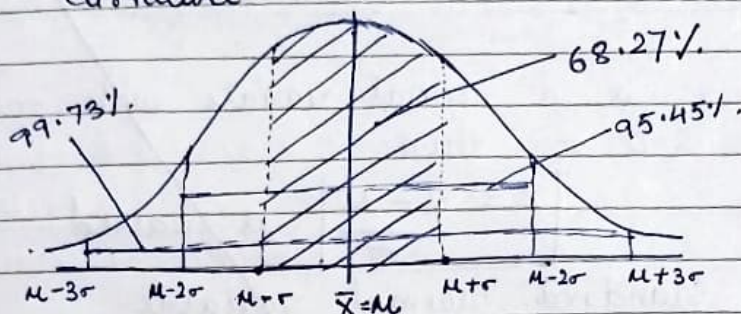
$$\Rightarrow \boxed{x = \mu \pm \sigma}$$

Date _____
Page _____

(iv) The point of inflexion of the curve are where $f''(x) = 0$, are

$$\boxed{x = \mu \pm \sigma}$$

ie the points where the $f(x)$ changes curvature.



probability that random variable lies in the interval $(\mu-\sigma, \mu+\sigma)$ is given by

$$P(\mu-\sigma < X < \mu+\sigma) = \int_{\mu-\sigma}^{\mu+\sigma} f(x) dx$$

$$= 0.6826$$

ie $\mu \pm \sigma$, covers 68.27% of the area under the curve.

Similarly $\mu \pm 2\sigma$ covers 95.45% Area.

$$\text{ie } \boxed{P(\mu-2\sigma < X < \mu+2\sigma) = 0.9545}$$

& $\mu \pm 3\sigma$ covers 99.73% Area

$$\text{ie } \boxed{P(\mu-3\sigma < X < \mu+3\sigma) = 0.9973}$$

(v)

The probability of getting a value between two no's is given by the area under the curve

$$P(x_1 < X < x_2) = \int_{x_1}^{x_2} f(x) dx = \frac{1}{\sigma\sqrt{2\pi}} \int_{x_1}^{x_2} e^{-\frac{(x-\mu)^2}{2\sigma^2}} dx$$

(vi) Symbolically $\boxed{X \sim N(\mu, \sigma^2)}$

Standard Normal variate:-

If $X \sim N(\mu, \sigma^2)$ then standard normal variate is given by.

$$Z = \frac{X - \mu}{\sigma}$$

If mean $\mu = 0$ and s.d $\sigma = 1$ then
 $Z = X \sim N(0, 1)$

* If $X \sim N(\mu, \sigma^2)$, then the prob. that r.v X will lie between $X = \mu$ and $X = x_1$ is

$$P(\mu < X < x_1) = \int_{\mu}^{x_1} f(x) dx.$$

$$P(\mu < X < x_1) = \int_{\mu}^{x_1} \frac{1}{\sigma \sqrt{2\pi}} e^{-\frac{1}{2} \left(\frac{x - \mu}{\sigma} \right)^2} dx$$

put $Z = \frac{X - \mu}{\sigma}$

when $X = \mu$, $Z = 0$ $dz = \frac{dx}{\sigma}$

when $X = x_1$ $Z = \frac{x_1 - \mu}{\sigma} = z_1$ (say)

$$P(\mu < X < x_1) = P(0 < Z < z_1)$$

$$P(0 < Z < z_1) = \frac{1}{\sqrt{2\pi}} \int_0^{z_1} e^{-z^2/2} dz = \int_0^{z_1} \phi(z) dz$$

where $\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2}$, $-\infty < z < \infty$ is

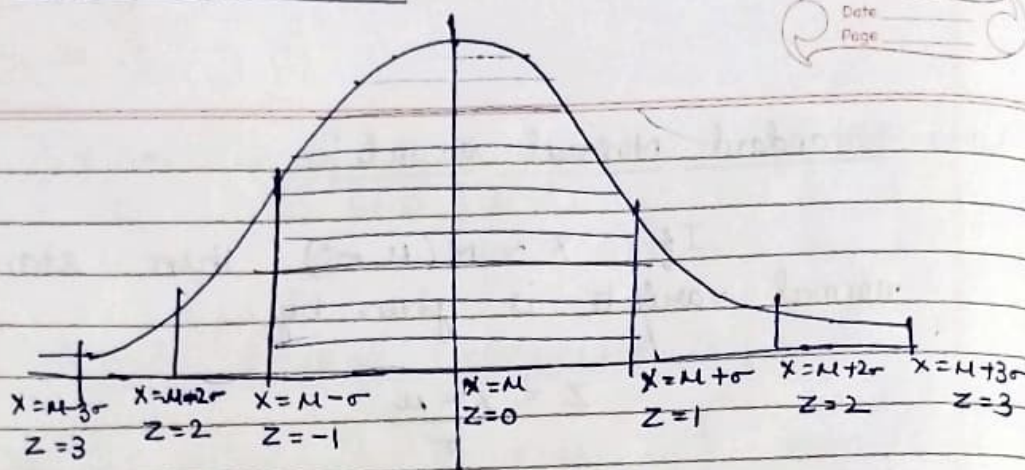
called probability function of standard normal variate.

Graph of $\phi(z)$:

classmate

Date

Page



Standard Normal distribution: $\phi(z)$ (P.d.f) :-

Hence the normal distribution $N(0,1)$ is called S.N.D, with density function.

$$\left[\phi(z) = \frac{1}{\sqrt{2\pi}} e^{-z^2/2} \quad -\infty < z < \infty \right]$$

where $Z \sim N(0,1)$ is standard normal variate.

Note:-

$$* P(\mu - \sigma < X < \mu + \sigma) = \int_{\mu - \sigma}^{\mu + \sigma} f(x) dx$$

$$P(-1 < Z < 1) = \int_{-1}^1 \phi(z) dz = 2 \int_0^1 \phi(z) dz \quad (\text{Symmetry})$$

$$= 2 \times 0.2420 = 0.4840$$

$$* P(\mu - 2\sigma < X < \mu + 2\sigma) = P(-2 < Z < 2) = 2 \int_0^2 \phi(z) dz$$
$$= 2 \times 0.4772 = 0.9544$$

$$* P(\mu - 3\sigma < X < \mu + 3\sigma) = P(-3 < Z < 3)$$
$$= 2 \int_0^3 \phi(z) dz = 2 \times 0.49865$$
$$= 0.9973$$

* Now prob that x lies outside the range $\mu \pm 3\sigma$ is given by

$$P(\mu + \sigma < x < \mu - 3\sigma) = P(3\sigma < x - \mu < -3\sigma)$$

i.e.

$$P(|x - \mu| > 3\sigma) = P(|Z| > 3)$$

$$= 1 - P(-3 \leq Z \leq 3)$$

$$= 1 - 0.9973$$

$$\boxed{P(|Z| \geq 3) = 0.0027}$$

Note

$$\int_{-\infty}^{\infty} f(x) dx = \int_{-\infty}^{\infty} \phi(z) dz = 1$$

Area under the normal probability curve is unity.

Q1 Assume mean height of soldiers to be 68.22 inches with a variance of 10.8 inches. How many soldiers in a regiment of 1000 would you expect is to be over 6 feet tall.

Solⁿ:-

$$N = 1000, \mu = 68.22, \sigma^2 = 10.8$$

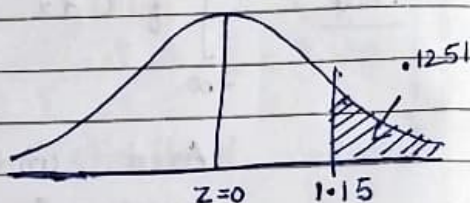
$$6 \text{ feet} = 72 \text{ inches}$$

$$\sigma = 3.286$$

X is a random variable denote the height of soldiers

$$Z = \frac{X - \mu}{\sigma}$$

$$Z = \frac{X - 68.22}{3.286}$$



at $X = 6$

$$Z = \frac{72 - 68.22}{3.286} = 1.15$$

ie to find ~~$P(X > 6)$~~ $P(Z > 1.15)$

$$= 0.5 - P(0 < Z < 1.15)$$

$$= 0.5 - 0.3749$$

$$= 0.1251$$

out of 1000 soldiers over six feet.
 $= 1000 \times 0.1251$

$$= 125.1 \approx \underline{125 \text{ soldiers}}$$

Q. X is normally distributed and mean of X is 12 and S.D is 4. Find the following probabilities

(a) $P(X \geq 20)$ (ii) $P(X \leq 20)$ (iii) $P(0 \leq X \leq 12)$

(b) Find x' when $P(X > x') = 0.24$

(c) Find x_0' and x_1' when $P(x_0' < X < x_1') = 0.50$ and $P(X > x_1') = 0.25$.

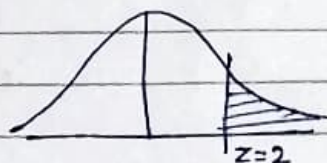
Solⁿ:-

mean $\mu = 12$, $\sigma = 4$.

define $Z = \frac{X - \mu}{\sigma} = \frac{X - 12}{4}$

(i) $P(X \geq 20)$

when $X = 20$, $Z = 2$



ie $P(X \geq 20) = P(Z \geq 2)$

$= 0.5 - P(0 \leq Z \leq 2)$

$= 0.5 - 0.4772$

$P(X \geq 20) = 0.0228$

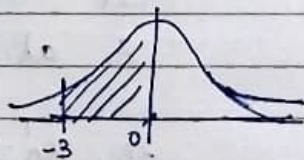
(ii) $P(X \leq 20) = 1 - P(X \geq 20) = 1 - 0.0228$
 $= 0.9772$

$P(X \leq 20) = 0.9772$

(iii) $P(0 \leq X \leq 12)$

when $X = 0$, $Z = -3$

$X = 12$, $Z = 0$



ie $P(0 \leq X \leq 12) = P(-3 \leq Z \leq 0)$
 $= P(0 \leq Z \leq 3)$ (from sym)
 $= 0.4987$

(b) find x_1' , when $P(X > x_1') = 0.24$
 when $X \geq x_1$, say $Z = z_1$

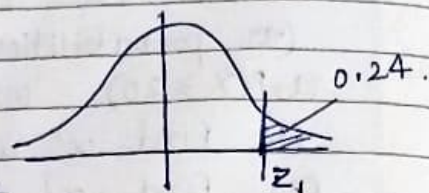
$$\text{ie } P(X > x_1) = P(Z > z_1) = 0.24$$

$$\text{ie } P(0 < Z < z_1) = 0.5 - 0.24$$

$$P(0 < Z < z_1) = 0.26$$

$$z_1 = 0.71 \text{ (approx)}$$

(from Tables)



$$\text{ie } \frac{X - \mu}{\sigma} = Z$$

$$\text{ie } \frac{x_1 - \mu}{\sigma} = z_1$$

$$\frac{x_1 - 12}{4} = 0.71$$

$$x_1 = 12 + 0.71 \times 4$$

$$\boxed{x_1 = 14.84}$$

$$\textcircled{c} \quad P(x_0' < X < x_1') = 0.50$$

$$P(X > x_1') = 0.25$$

$$\text{as } \frac{X - \mu}{\sigma} = Z$$

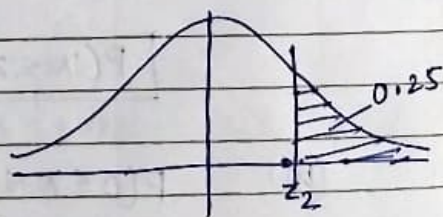
$$\text{let at } X = x_0', \quad Z = z_1$$

$$\& \text{ at } X = x_1', \quad Z = z_2$$

$$\text{ie } \frac{x_0' - \mu}{\sigma} = z_1, \quad \frac{x_1' - \mu}{\sigma} = z_2$$

$$\text{ie } P(z_1 < Z < z_2) = 0.50$$

$$\& \quad P(X > z_2) = 0.25$$



ie from figure its clear that $z_1 = -z_2$

$$\text{ie } P(-z_2 < Z < z_2) = 0.50$$

$$P(Z > z_2) = 0.25$$

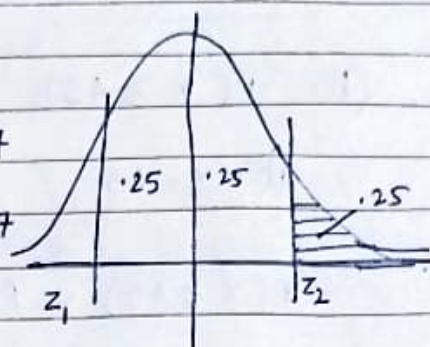
$$\text{ie } P(0 < Z < z_2) = 0.25$$

From table

$$z_2 = 0.67, z_1 = -0.67$$

$$\frac{x'_1 - \mu}{\sigma} = z_2 \Rightarrow x'_1 = 12 + 4 \times 0.67$$

$$x'_1 = 14.68$$



$$\text{Q. 8 } \frac{x'_0 - \mu}{\sigma} = -0.67$$

$$x'_0 = 12 - 4 \times 0.67$$

$$x'_0 = 9.32$$

Q: X is a Normal variate with mean 30 and S.D 5, Find the probabilities that

$$(i) 26 \leq X \leq 40 \quad (ii) X \geq 45 \quad (iii) |X - 30| > 75$$

Soln: Say Z is standard normal variate

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 30}{5}$$

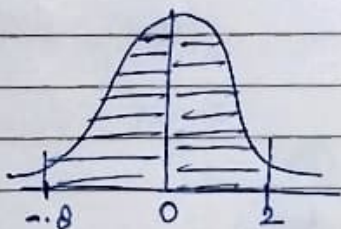
$$\text{when } X = 26, \quad Z = \frac{26 - 30}{5} = -0.8$$

$$\text{at } X = 40, \quad Z = \frac{40 - 30}{5} = 2$$

$$\therefore P(26 \leq X \leq 40) = P(-0.8 \leq Z \leq 2)$$

$$= P(-0.8 \leq Z \leq 0) + P(0 \leq Z \leq 2)$$

$$= P(0 \leq Z \leq 0.8) + P(0 \leq Z \leq 2) \quad (\text{By symmetry})$$



from Table

$$P(26 \leq X \leq 40) = 0.2881 + 0.4772$$

$$P(26 \leq X \leq 40) = 0.7653$$

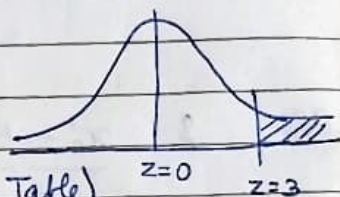
$$(b) P(X \geq 45)$$

$$\text{at } x = 45, \quad z = \frac{45 - 30}{5} = 3$$

$$P(X \geq 45) = P(Z \geq 3)$$

$$= 0.5 - P(0 \leq Z \leq 3)$$

$$= 0.5 - 0.4987 \quad (\text{From Table})$$



$$P(X \geq 45) = 0.0044$$

$$(c) P(|X - 30| > 5) = 1 - P(|X - 30| \leq 5)$$

$$= 1 - P(25 \leq X \leq 35)$$

①

$$P(|X - 30| \leq 5) = P(-5 \leq X - 30 \leq 5)$$

$$= P(25 \leq X \leq 35)$$

$$= P(-1 \leq Z \leq 1)$$

$$= 2 P(0 \leq Z \leq 1)$$

$$= 2 \times 0.2420$$

$$\text{ie } P(|X - 30| \leq 5) = 0.4840$$

$$\text{ie } P(|X - 30| > 5) = 1 - 0.4840$$

$$= 0.5160$$

(from ①)

Since
 $|X| < a$
 $\Rightarrow -a < X < a$

Q: In a distribution exactly normal, 10.03% of items are under 25 kilogram and 89.97% of the items are under 70 kilogram weight. What are mean and S.D of the distribution.

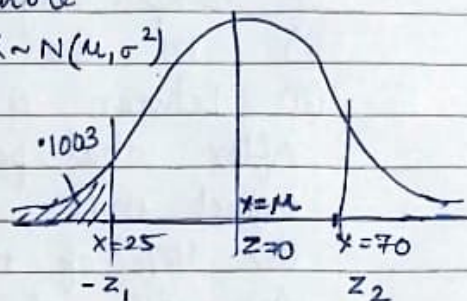
solⁿ: Let X is the r.v denote the weight of items, & $X \sim N(\mu, \sigma^2)$

• given that

$$P(X < 25) = .1003$$

and

$$P(X < 70) = .8997$$



• say Z is standard Normal variate
s.t $Z \sim N(0,1)$

$$Z = \frac{X - \mu}{\sigma}$$

ie ~~Rewrite~~ at $x = 25$, $Z = \frac{25 - \mu}{\sigma} = -z_1$ (say)

at $x = 70$, $Z = \frac{70 - \mu}{\sigma} = z_2$ (say)

ie $P(Z < -z_1) = .1003$ & $P(Z < z_2) = .8997$

$$.5 - P(-z_1 < Z < 0) = .1003$$

$$.5 - P(0 < Z < z_1) = .1003$$

$$\text{ie } P(0 < Z < z_1) = .5 - .1003 = .3997$$

from Normal Table

$$\boxed{z_1 = 1.28}$$

$$P(Z < z_2) = 0.5 + P(0 < Z < z_2)$$

$$= .8997$$

$$\text{ie } P(0 < Z < z_2) = .3997$$

From Normal Table

$$\boxed{z_2 = 1.28}$$

$$\text{ie } -z_1 = \frac{25 - \mu}{\sigma} \quad \& \quad z_2 = \frac{70 - \mu}{\sigma}$$

$$-1.28 = \frac{25 - \mu}{\sigma} \quad \& \quad 1.28 = \frac{70 - \mu}{\sigma} \quad \text{--- (1) \quad (2)}$$

On solving

$$\boxed{\sigma = 17.578}$$

$$\boxed{\mu = 47.5}$$

- Q: The local authorities in a certain city instal 10,000 electric lamps in the street of the city. If these lamps have an average life of 1,000 burning hours with a s.d of 200 hours, assuming normality, what no. of lamps might be expected to fail
- in the first 800 burning hours?
 - between 800 and 1200 burning hours?
- After what period of burning hours would you expect that
- 10% of the lamps would fail
 - 10% of the lamps would be still burning.

Solution: $X \sim N(\mu, \sigma^2)$ is a normal variate with
 mean = 1000, & s.d = 200 hours, denotes the life a bulb.
 $N = 10,000$

$$Z = \frac{X - \mu}{\sigma} = \frac{X - 1000}{200}$$

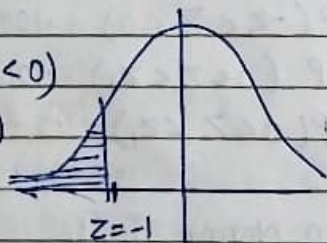
(i) The probability that bulb fails in the first 800 burning hours is given by

$$P(X < 800) = P(Z < -1)$$

$$= 0.5 - P(-1 < Z < 0)$$

$$= 0.5 - P(0 < Z < 1)$$

$$= 0.5 - 0.3413$$



$$P(X < 800) = 0.1587$$

(Ans) Out of 10,000 bulbs, no. of bulbs which fails in the first 800 hours
 $= 10,000 \times 0.1587$
 $= 1587$ Ans.

$$\begin{aligned}
 \text{(ii)} \quad P(800 < X < 1200) &= P(-1 < Z < 1) \\
 &= 2 P(0 < Z < 1) \\
 &= 2 \times 0.3413 \\
 &= 0.6826
 \end{aligned}$$

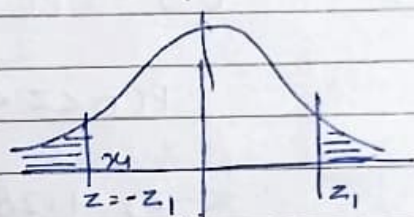
Hence the expected no. of bulbs with life b/w 800 and 12,00 hours of burning life is

$$10,000 \times 0.6826 = 6,826$$

(a) Let 10% of bulbs fails after x_1 hours of burning life. then to find x_1 such that

$$P(X < x_1) = 0.10$$

$$Z = \frac{X - \mu}{\sigma}$$



$$\text{i.e. } Z = \frac{x_1 - 1000}{200} = -z_1, \text{ say}$$

$$P(Z < -z_1) = 0.10$$

$$P(Z > z_1) = 0.10$$

$$\text{i.e. } P(0 < Z < z_1) = 0.50 - 0.10$$

$$P(0 < Z < z_1) = 0.40$$

from Normal Table

$$\boxed{z_1 = 1.28}$$

$$\text{i.e. } -1.28 = \frac{x_1 - 1000}{200}$$

$$x_1 = 1000 - 1.28 \times 200$$

$$x_1 = 1000 - 256$$

$$\boxed{x_1 = 744}$$

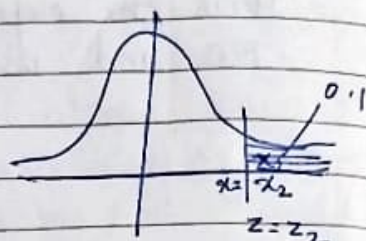
After 744, hours of burning life 10% of bulb will fail.

(b) ~~After~~ After what period of burning hours 10% of the ~~burn~~ lamps would be still burning?

ie say after x_2 hours, 10% of lamps would be still burning

say $P(X > x_2) = 0.1$

say $z_2 = \frac{x_2 - 1000}{200}$



ie $P(Z > z_2) = 0.1$

$0.5 - P(0 < Z < z_2) = 0.1$

$P(0 < Z < z_2) = 0.40$

$z_2 = 1.28$ (from table)

ie $z_2 = \frac{x_2 - 1000}{200}$

$1.28 = \frac{x_2 - 1000}{200}$

$x_2 = 1256 \text{ hours}$ Ans

After 1,256 hours of burning life, 10% of bulb will be still burning.

Q1. A grinding machine is so set that its production of shafts has an average diameter of 10.10 cm and S.D is 0.20 cm. The product specifications call for shaft diameters b/w 10.05 cm and 10.20 cms. What proportion of output meets the specifications presuming normal distribution?

Ans: 29.02%

Q) In a sample of 1000 cases, the mean of certain test is 14, & S.D is 2.5. Assuming the distribution to be normal, find.

(i) how many students score b/w 12 and 15.

(ii) " " " " above 18.

(iii) " " " " below 18.

Solution $N=1000$, $\mu=14$, $\sigma=2.5$

X is r.v denotes marks obtain by students

(a) $P(12 < X < 15)$

$$\text{say } Z = \frac{X - \mu}{\sigma} = \frac{X - 14}{2.5}$$

$$\text{at } X=12, Z_1 = -0.8$$

$$\text{at } X=15, Z_2 = 0.4$$

$$\Rightarrow P(12 < X < 15) = P(-0.8 < Z < 0.4)$$

$$= P(-0.8 < Z < 0) + P(0 < Z < 0.4)$$

$$= P(0 < Z < 0.8) + P(0 < Z < 0.4)$$

(by symm)

$$= 0.2882 + 0.1554$$

$$= 0.4435$$

No of student scores b/w 12 & 15

$$= N \times P(12 < X < 15)$$

$$= 1000 \times 0.4435 = 444$$

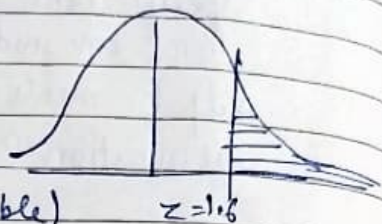
$$(2) P(X > 18)$$

$$\text{at } x=18, z = \frac{18-14}{2.5} = 1.6$$

$$P(X > 18) = P(Z > 1.6)$$

$$= 0.5 - P(0 < Z < 1.6)$$

$$= 0.5 - 0.4452 \text{ (from Table)}$$



$$P(X > 18) = 0.0548$$

$$\text{No. of Student} = 1000 \times 0.0548$$

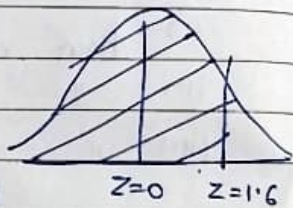
$$[\text{No. of Students} = 55]$$

$$(3) P(X < 18) = P(Z < 1.6)$$

$$= 0.5 + P(0 < Z < 1.6)$$

$$= 0.5 + 0.4452$$

$$= 0.9452$$



$$\text{No. of Students} = 1000 \times 0.9452$$

$$= 945 \quad \underline{\text{Ans.}}$$